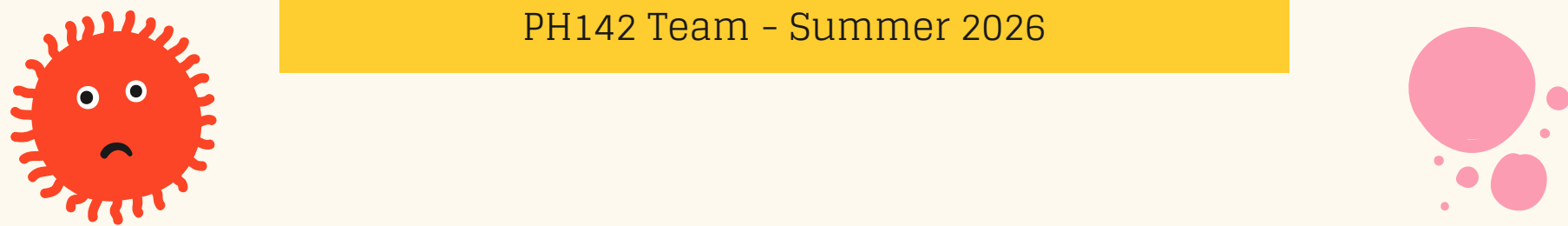




Week 4 Review

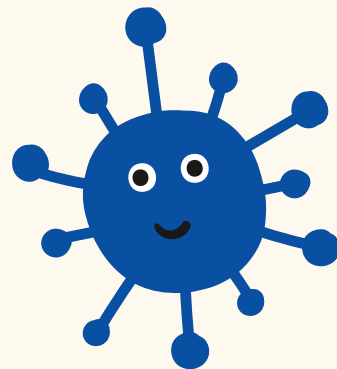
Probability, Power, Error, Sample Size



PH142 Team - Summer 2026



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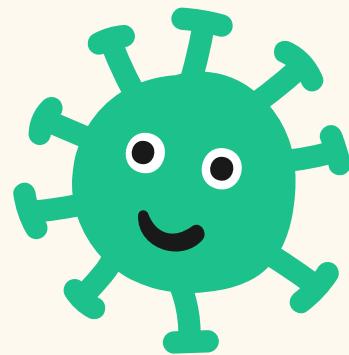
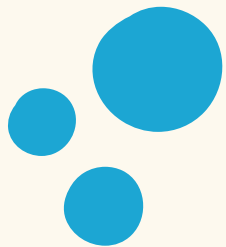
Normal, Binomial,
Poisson

04

Inference

Confidence intervals (known
SD), Hypothesis testing, Power,
type I and type II error, sample
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Probability



What is Probability?

- **What is it?** Probability is the language we use to describe uncertainty. It quantifies the chance of a specific event happening. We can think of it as the
- **long-run relative frequency** of an event—if we repeated an experiment over and over, the proportion of times the event occurs would get closer and closer to its true probability.
- **Why does it matter?** From interpreting medical test results to understanding election polls, probability helps us make sense of a world full of randomness and incomplete information.
- **Fundamental Components:** Every probability problem involves three core ideas:
 - A **random experiment**: An action or process with an uncertain outcome (e.g., flipping a coin, rolling a die, taking a random sample).
 - The
 - **sample space (S)**: The set of *all possible outcomes* of the experiment. For a coin flip, $S = \{\text{Heads}, \text{Tails}\}$. For a six-sided die, $S = \{1, 2, 3, 4, 5, 6\}$.
 - An **event (A)**: A specific outcome or a collection of outcomes that we are interested in (e.g., getting heads, rolling an even number).



The Law of Large Numbers in Action

- The "long-run frequency" idea is formalized by the **Law of Large Numbers**. This law guarantees that as you increase your sample size, your sample estimate (like a proportion) will get closer to the true population parameter.
- **Illustrative Example: The Common Cold**
 - In your lecture, you saw an example where a population had a true cold prevalence of 11.2%.
 - When a tiny sample of 5 people was taken, the sample estimate was 0%—not very accurate!
 - As the sample size increased to 1,000, 10,000, and finally the whole population, the sample proportion got progressively closer to the true value of 11.2%.
 - This shows that
 - **larger samples give us more precise estimates**, but we have to be mindful of sampling bias!



The Law of Large Numbers in Action

- R Simulation: Flipping a Coin
- We can simulate this idea in R. Let's flip a virtual coin 5000 of times and watch the cumulative proportion of heads converge to the true probability of 0.5.

```
##{r}
# Load necessary library
library(ggplot2)
library(dplyr)

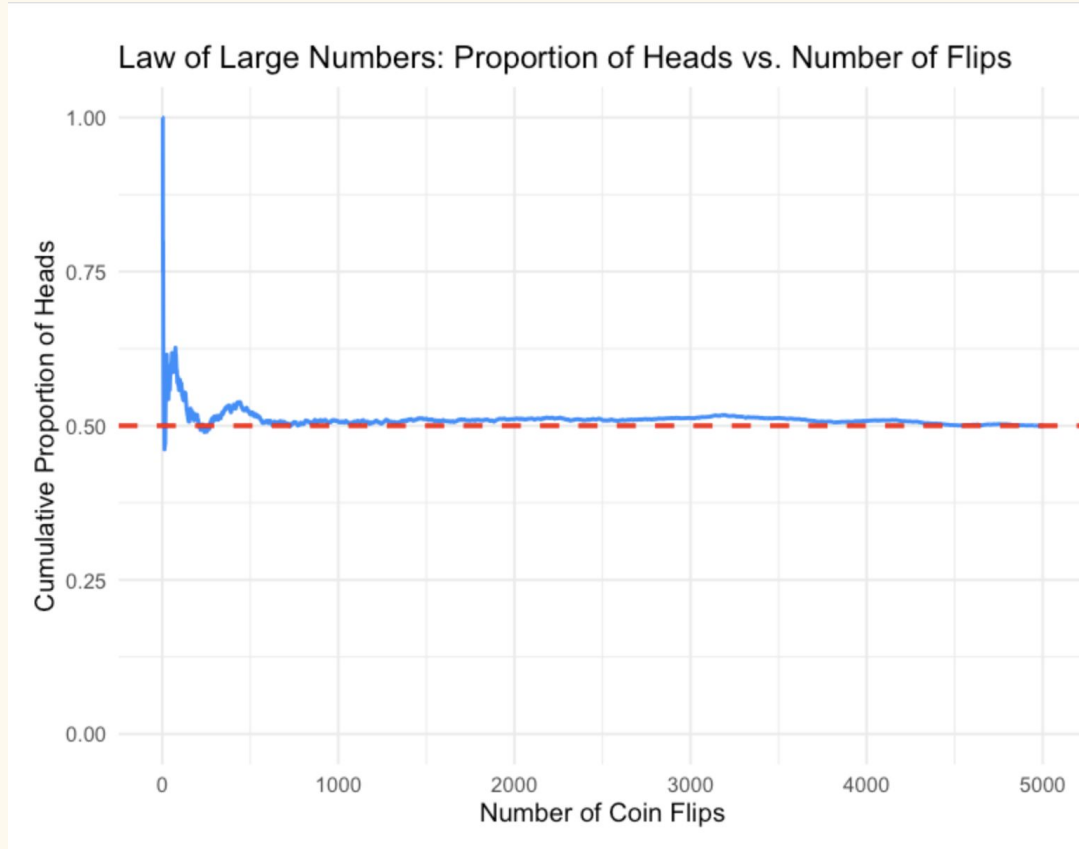
# Set up the simulation
set.seed(123) # for reproducibility
n_flips <- 5000
coin_flips <- sample(c("Heads", "Tails"), n_flips, replace = TRUE)

# Calculate the cumulative proportion of heads
results <- data.frame(
  flip_number = 1:n_flips,
  outcome = coin_flips
) %>%
  mutate(
    is_heads = ifelse(outcome == "Heads", 1, 0),
    cumulative_heads = cumsum(is_heads),
    cumulative_prop = cumulative_heads / flip_number
  )

# Plot the results
ggplot(results, aes(x = flip_number, y = cumulative_prop)) +
  geom_line(color = "dodgerblue", linewidth = 1) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color = "red", linewidth = 1.2) +
  labs(
    title = "Law of Large Numbers: Proportion of Heads vs. Number of Flips",
    x = "Number of Coin Flips",
    y = "Cumulative Proportion of Heads"
  ) +
  ylim(0, 1) + # Set y-axis limits from 0 to 1
  theme_minimal(base_size = 14)
##
```



The Law of Large Numbers in Action



Fundamental Probability Rules



Rule 1: The Range of Probability

- The probability of any event, $P(A)$, must be between 0 and 1, inclusive. $0 \leq P(A) \leq 1$.
- A probability of 0 means the event is impossible.
- A probability of 1 means the event is certain.

Rule 2: The Sample Space Rule

The sum of the probabilities of all possible outcomes in the sample space must equal 1. If you've listed every possibility, one of them *must* happen.



Fundamental Probability Rules



* Rule 3: The Complement Rule

- The probability that an event A does *not* happen is denoted $P(A^c)$ or $P(\bar{A})$.
- This probability is calculated as $P(A^c) = 1 - P(A)$.
- This is incredibly useful! Sometimes it's much easier to calculate the probability of an event *not* happening and then subtract from 1.

Example Question:

The probability of rain tomorrow is 40%. What is the probability that it will *not* rain?

Answer: $P(\text{no rain}) = 1 - P(\text{rain}) = 1 - 0.40 = 0.60$, or 60%



Combining Events & Venn Diagrams

- Often, we're interested in the probability of more than one event.
- **Composite Events:** These are events that involve the outcomes of two or more other events, A and B.
 - **Intersection ($A \cap B$):** Represents the event that **both A AND B occur**. In a Venn diagram, this is the overlapping area of the two circles.
 - **Union ($A \cup B$):** Represents the event that **A OR B (or both) occur**. In a Venn diagram, this is the total area covered by both circles, including their overlap.



Combining Events & Venn Diagrams

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Combining Events & Venn Diagrams

- Our lecture slides show a Venn Diagram for survey data on vaping and JUUL ad exposure.
- There are two overlapping circles:
 - **Circle J:** Represents the 18% of people who saw a JUUL ad ($P(J)=0.18$).
 - **Circle V:** Represents the 11% of people who vaped in the last 30 days ($P(V)=0.11$).
 - **The Overlap:** Represents the 3.6% who did both ($P(J \cap V)=0.036$).



Combining Events & Venn Diagrams

* Let's represent this in a 2x2 table instead of a diagram:

```
# Load the knitr library for clean tables
library(knitr)

# Create the data for the table from the lecture example
vape_data <- matrix(c(0.036, 0.144, 0.074, 0.746), ncol = 2, byrow = TRUE)
colnames(vape_data) <- c("Vaped", "Did Not Vape")
rownames(vape_data) <- c("Saw JUUL Ad", "Did Not See Ad")

# Add margins (totals)
vape_table <- addmargins(vape_data)

# Print the table
kable(vape_table, caption = "Probability Table for Vaping and JUUL Ad Exposure")
```

Probability Table for Vaping and JUUL Ad Exposure

	Vaped	Did Not Vape	Sum
Saw JUUL Ad	0.036	0.144	0.18
Did Not See Ad	0.074	0.746	0.82
Sum	0.110	0.890	1.00

THE GENERAL ADDITION RULE



- This rule tells us how to calculate the probability of a **union** of two events.
 - **The Formula:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Why do we subtract the intersection?
 - If we simply add $P(A)$ and $P(B)$ we **double-count** the overlapping area (the intersection, $P(A \cap B)$). The subtraction corrects for this.
- **Special Case: Disjoint Events**
 - If two events A and B are **disjoint** (mutually exclusive), they have no overlap. Their intersection is 0, so $P(A \cap B) = 0$
 - In this case, the formula simplifies to: $P(A \cup B) = P(A) + P(B)$



THE GENERAL ADDITION RULE



- **Example Question:** Using the JUUL/Vaping data, what is the probability that a randomly chosen person either saw a JUUL ad or vaped in the last 30 days?
- **Answer:** We are looking for $P(J \cup V)$. Using the formula: $P(J \cup V) = P(J) + P(V) - P(J \cap V) = 0.18 + 0.11 - 0.036 = 0.254$. So, there is a 25.4% chance that a person vaped or saw a JUUL ad.



Conditional Probability



- **Definition:** Conditional probability is the likelihood of an event (B) occurring, **given that** another event (A) is known to have already occurred. This concept is fundamental for understanding how new information can change our assessment of probability.
- **The Big Idea:** Instead of looking at the entire sample space, we **reduce our sample space** to only the outcomes where event A happened. Then, within that new, smaller world, we calculate the probability of B.
- **Notation:** We write it as $P(B|A)$. This is read as "the probability of B, given A."
- **The Formula:** The probability of B given A is the probability of both A and B happening, divided by the probability of A happening:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$



Conditional Probability



- Our lecture slides use the example of shark attacks to question broad probability claims.
- The overall probability of a person dying from a shark attack is incredibly low (e.g., 1 in 3.7 million). However, what is the probability of a shark attack, **given that you are swimming in the ocean**?
 - This is a conditional probability! The condition ("swimming in the ocean") dramatically reduces the sample space from "all people" to "people currently in the ocean," making the probability much higher and more relevant.





The General Multiplication Rule

- By rearranging the conditional probability formula, we get the **General Multiplication Rule**. This rule is used to find the probability of two events happening together (their intersection).

- **Formula:** The probability that two events A and B occur is given by:

$$P(A \cap B) = P(A) \times P(B|A)$$

- **In Words:** The chance of A and B both happening is the chance that A happens, multiplied by the chance that B will happen *given that A has already happened*.

- **When to Use It**

- This rule is essential for calculating probabilities in multi-stage events, like drawing two cards from a deck without replacement.

- **Example:** What is the probability of drawing two Kings from a standard 52-card deck?

- Let A = “first card is a King.” $P(A \cap B) = P(A) \times P(B|A) = \frac{4}{52} \times \frac{3}{51} \approx 0.0045$ and card is a King.” $P(A)=4/52$.

- *Given* the first was a king, there are now only 51 cards left, and only 3 of

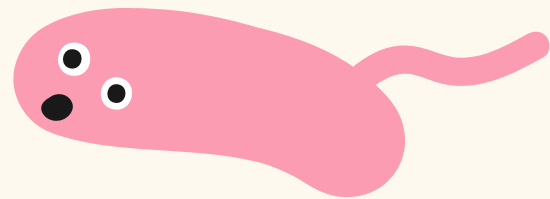
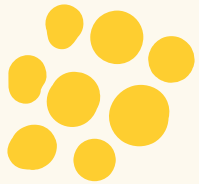


INDEPENDENT VS. DEPENDENT EVENTS



- The concept of independence is formally defined by conditional probability.
- **Independent Events:** Two events are **independent** if knowing that one event occurred does not change the probability that the other occurred.
 - **Formal Test:** Events A and B are independent if any of these (equivalent) conditions are true:
 - $P(B|A) = P(B)$
 - $P(A|B) = P(A)$
 - $P(A \cap B) = P(A) * P(B)$ (This is the simplified multiplication rule for independent events)
 - **Example:** The outcome of one coin flip is independent of the next.
- **Dependent Events:** Two events are **dependent** if knowing one event occurred *does* change the probability of the other.
 - **Formal Test:** If $P(B|A) \neq P(B)$, the events are dependent.
 - **Example:** Drawing a card from a deck and *not* replacing it. The probability of the second draw depends on the result of the first.





02

Distributions

Normal, Binomial, Poisson





The Binomial Distribution

The Binomial Distribution

The Binomial distribution is our go-to model for **discrete** data when we're counting the number of "successes" in a fixed number of trials.

The Binomial Setting: To use this distribution, the situation must meet four conditions, known as a **Bernoulli Process**:

1. There is a **fixed number of trials** or observations, which we call **n**.
2. Each of the **n** observations is **independent**. [cite_start]Knowing the outcome of one trial doesn't change the probability of the others[cite: 6651].
3. Each observation falls into one of just **two categories**, which we label "success" and "failure" for convenience.
4. The probability of a "success," which we call **p**, is the **same for each observation**.

- **Notation:** $X \sim B(n, p)$.
- **Mean and Standard Deviation:**
- Mean: $\mu = np$ (The expected number of successes).
- Standard Deviation: $\sigma = \sqrt{np(1-p)}$

Example Scenarios:

- The number of heads in 10 coin flips ($n = 10, p = 0.5$).
- The number of patients who recover out of 250 given a new treatment ($n = 250, p = \text{recovery rate}$).
- The number of correctly answered questions on a 20-question multiple-choice test by guessing randomly.



The Binomial Distribution



Binomial Probabilities & R Functions

Calculating Binomial Probabilities: We can find the probability of getting exactly k successes in n trials using the following formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- The term $\binom{n}{k}$ is the **binomial coefficient**, read as “n choose k.” It calculates how many different ways you can arrange k successes among n trials.

Using R for Binomial Probabilities

Calculating this by hand can be tedious. R has functions to make it easy:

- `dbinom(x, size, prob)` : Calculates the probability of an **exact** number of successes. Use it to find $P(X = k)$.
- `pbinom(q, size, prob)` : Calculates the **cumulative** probability. Use it to find $P(X \leq k)$.

Illustrative Graph: Binomial Probability Distribution

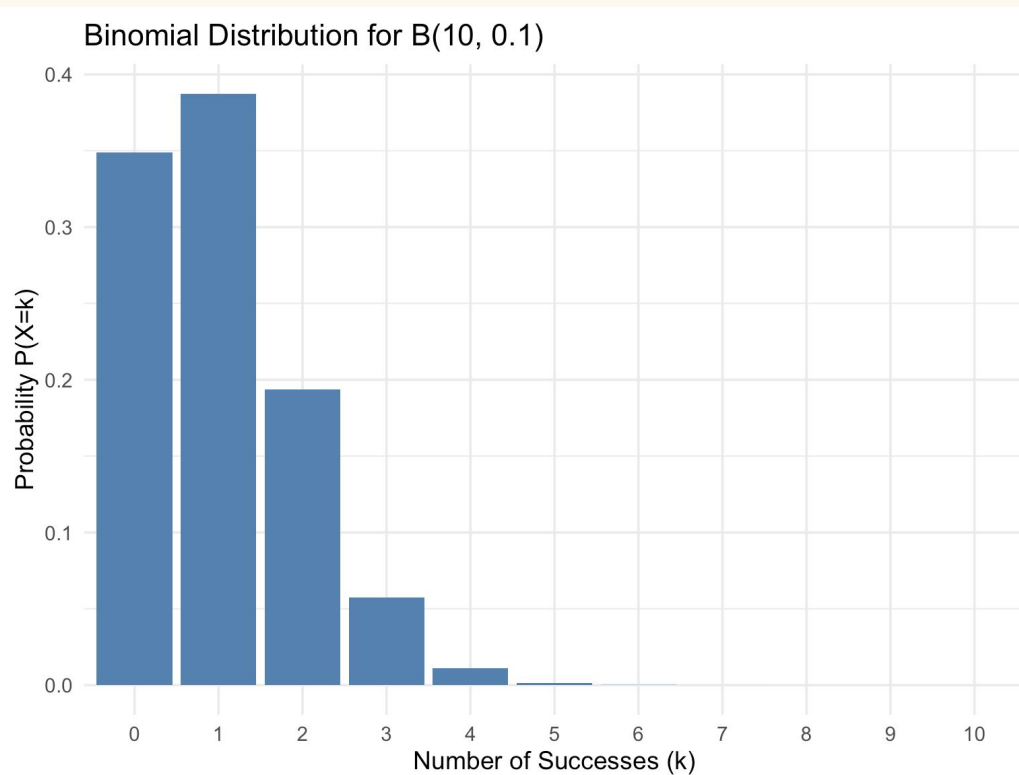
This plot shows the probability of each possible outcome for our contaminated bottle example, where $X \sim B(10, 0.1)$. Notice that 1 success is the most likely outcome, and the distribution is skewed to the right because p is low.

```
# Create data for the plot
successes <- 0:10
probabilities <- dbinom(x = successes, size = 10, prob = 0.1)
binom_data <- data.frame(successes, probabilities)

# Plot the distribution
ggplot(binom_data, aes(x = factor(successes), y = probabilities)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Binomial Distribution for B(10, 0.1)",
    x = "Number of Successes (k)",
    y = "Probability P(X=k)"
  ) +
  theme_minimal(base_size = 14)
```



The Binomial Distribution



The Normal Distribution: The Bell Curve



- **Definition:** The Normal distribution is the most famous distribution in statistics. It's used for **continuous** variables (like height, weight, or temperature) whose distributions are **symmetric**, **unimodal**, and **bell-shaped**. Many natural phenomena follow this pattern, which makes it an incredibly useful model.
- **Key Parameters:** A Normal distribution is completely defined by two parameters:
 - **Mean (μ):** This is the center of the distribution. It determines where the peak of the bell curve is located.
 - **Standard Deviation (σ):** This measures the spread or variability of the data. A smaller σ results in a taller, narrower curve, while a larger σ results in a shorter, wider curve.
 - **Notation:** We write that a variable X follows a Normal distribution as $X \sim N(\mu, \sigma)$.



Visualizing Normal Distribution

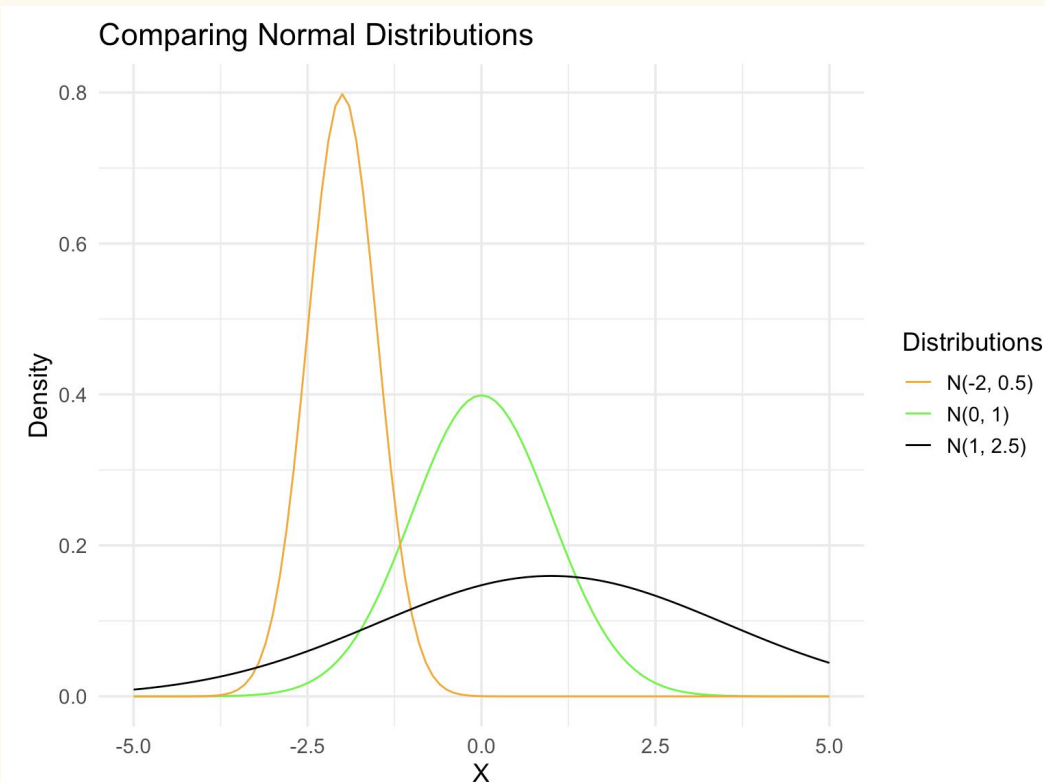
This plot shows how changing the mean and standard deviation affects the shape of the curve.

- The mean (μ) shifts the curve left or right.
- The standard deviation (σ) controls the spread. Notice the green curve ($sd = 0.5$) is much narrower and taller than the black curve ($sd = 2.5$). This is because the total area under each curve must equal 1.

```
# Load necessary library
library(ggplot2)
library(dplyr)

# Create a plot showing multiple Normal distributions
ggplot(data.frame(x = c(-5, 5)), aes(x = x)) +
  stat_function(fun = dnorm, n = 101, args = list(mean = 0, sd = 1), aes(color = "N(0, 1)")) +
  stat_function(fun = dnorm, n = 101, args = list(mean = -2, sd = 0.5), aes(color = "N(-2, 0.5)")) +
  stat_function(fun = dnorm, n = 101, args = list(mean = 1, sd = 2.5), aes(color = "N(1, 2.5)")) +
  scale_color_manual("Distributions", values = c("orange", "green", "black")) +
  labs(
    title = "Comparing Normal Distributions",
    x = "X",
    y = "Density"
  ) +
  theme_minimal(base_size = 14)
```

Visualizing Normal Distribution





The 68-95-99.7 Rule

- This rule gives you a quick way to approximate probabilities for any Normal distribution without complex calculations.
- Key observations:
 - Approximately **68%** of the observations fall within **1 standard deviation** of the mean ($\mu \pm 1\sigma$).
 - Approximately **95%** of the observations fall within **2 standard deviations** of the mean ($\mu \pm 2\sigma$).
 - Approximately **99.7%** of the observations fall within **3 standard deviations** of the mean ($\mu \pm 3\sigma$).

•





Z-Scores

- A Z-score is a standardized value that tells you how many standard deviations an observation (x) is from the mean (μ).
 - **Purpose:** Z-scores allow us to convert any Normal distribution into the **Standard Normal Distribution** ($N(0,1)$), which has a mean of 0 and a standard deviation of 1. This is useful for comparing values from different scales or finding probabilities using a standard table or software.
 - **Formula:**

$$Z = \frac{x - \mu}{\sigma}$$



Example Question



- **Context:** The distribution of heights of young women is approximately Normal, with a mean $\mu=64.5$ inches and a standard deviation $\sigma=2.5$ inches. So, $H \sim N(64.5, 2.5)$.
- **Question:** What percentage of young women are between 62 and 67 inches tall?
- **Answer using the 68-95-99.7 Rule:**
 - Notice that 62 inches is one standard deviation below the mean ($64.5 - 2.5$).
 - Notice that 67 inches is one standard deviation above the mean ($64.5 + 2.5$).
 - The range from 62 to 67 inches is therefore $\mu \pm 1\sigma$.
 - According to the rule, this range contains approximately **68%** of the observations.





Example Question

- **Context:** The distribution of heights of young women is approximately Normal, with a mean $\mu=64.5$ inches and a standard deviation $\sigma=2.5$ inches. So, $H \sim N(64.5, 2.5)$.
- **Question:** What percentage of young women are between 62 and 67 inches tall?
- **Answer using R Coding:**

```
pnorm(q = 67, mean = 64.5, sd = 2.5) - pnorm(q = 62, mean = 64.5, sd = 2.5)
```

```
## [1] 0.6826895
```





The Poisson Distribution

- The Poisson distribution is another model for **discrete** data. Instead of counting successes in a fixed number of trials (like Binomial), it counts the number of events that occur in a **fixed interval of time or space**.
- **When to use it?** Use the Poisson distribution when:
 1. You are counting the number of occurrences of an event in a fixed interval (e.g., a day, a square meter, a page of text).
 2. The occurrences are **independent** of one another.
 3. The probability of an occurrence is the **same for all equal-sized intervals**.
 4. There is **no theoretical upper limit** on the number of occurrences.
 5. **Key Difference from Binomial:** For a binomial, the maximum number of successes is n . For a Poisson, the count can theoretically go to infinity.



- **Notation:** $X \sim \text{Pois}(\lambda)$, where λ is the **average number of events** in that interval.
- **Mean and Standard Deviation:** The mean of a Poisson distribution is λ . The variance is also λ , which means the standard deviation is $\sigma = \sqrt{\lambda}$.





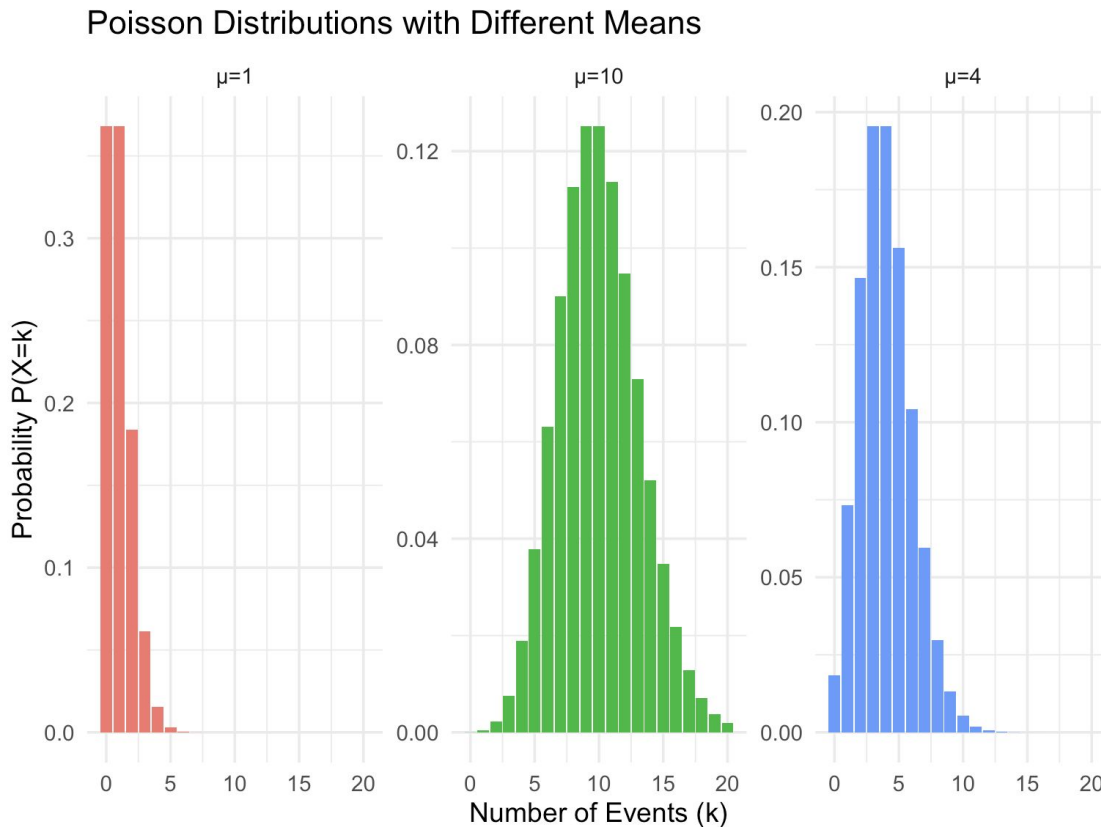
The Poisson Distribution

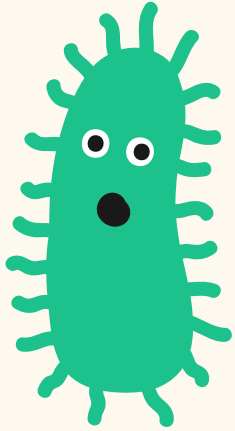
- **Example Scenarios:**

- The number of emails you receive in an hour.
 - The number of bacterial colonies growing on a petri dish.
 - The number of cars that pass through an intersection in a minute.
- The shape of a Poisson distribution depends on its mean, λ . When λ is small, the distribution is highly skewed right. As λ increases, the distribution becomes more symmetric and bell-shaped.



The Poisson Distribution





Sampling and CLT

03



Parameters vs Statistics

- **The Big Idea: Sample vs. Population**
- In statistics, we almost never have data for an entire **population**. Instead, we work with a smaller **sample** to make an educated guess, or **inference**, about that population. This distinction is crucial.
- **Parameter:** A number that describes a characteristic of the **population**.
 - It's a fixed, single value, but it's usually **unknown**.
 - *Examples:* The true mean height (μ) of all US women; the true proportion (p) of all California voters who support a certain policy.
- **Statistic:** A number that is computed from a **sample**.
 - We use statistics to **estimate** unknown parameters.
 - A statistic is a **random variable** because its value would change if we took a different random sample.
 - *Examples:* The mean height ($\bar{x}$) from a sample of 100 women; the proportion ($\hat{p}$) from a poll of 1500 voters.



Context: Why This Matters



- Imagine B and E both want to estimate the mean height of US women (μ). They each take a random sample of 100 women.
- B gets a sample mean of $\bar{x}=61.8$ inches.
- E gets a sample mean of $\bar{x}=64.4$ inches.
- Neither of their statistics is the true parameter μ , but they are estimates. The goal of inference is to understand how much these sample statistics vary and how we can use them to build a reliable range for the unknown parameter.





The Sampling Distribution

- The **sampling distribution** is the probability distribution of a statistic (like the sample mean, $\bar{x}$).
- It's a theoretical idea based on a thought experiment:
 - 1. Imagine taking **every possible random sample** of a specific size, n , from a population.
 - 2. Calculate your statistic of interest (e.g., the sample mean, $\bar{x}$) for each and every sample.
 - 3. Create a histogram of all those calculated statistics.
- The resulting distribution is the sampling distribution. In practice, we only take one sample, but understanding this theoretical distribution allows us to know how much our single sample's statistic is likely to vary from the true parameter.





Key Properties of the Sampling Distribution of the Mean

- **Center:** The mean of the sampling distribution is equal to the true population mean, μ . This means the sample mean, $\bar{x}$, is an **unbiased estimator** of μ .
- **Spread:** The standard deviation of the sampling distribution is called the **standard error (SE)**. It's smaller than the population standard deviation and is calculated as:

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

- This formula shows that as your **sample size (n) increases**, the **standard error decreases**. This means larger samples give you a more precise estimate of the population mean.





The Central Limit Theorem

- The Theorem

- For any population with a finite standard deviation σ , if the sample size n is sufficiently large, the sampling distribution of the sample mean ($\bar{x}$) will be approximately Normal.

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

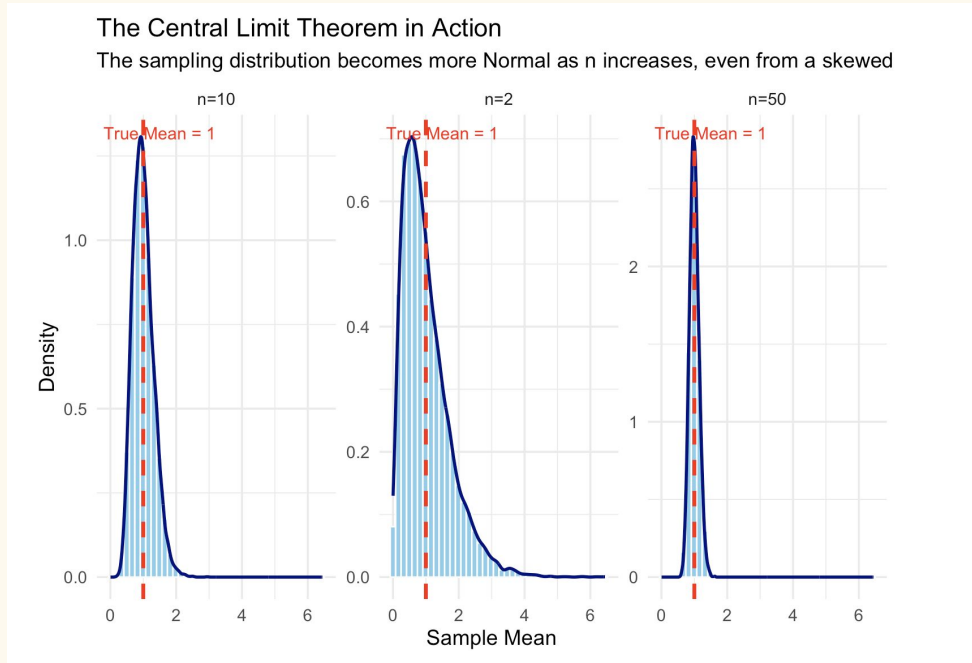
- Why is this so powerful?

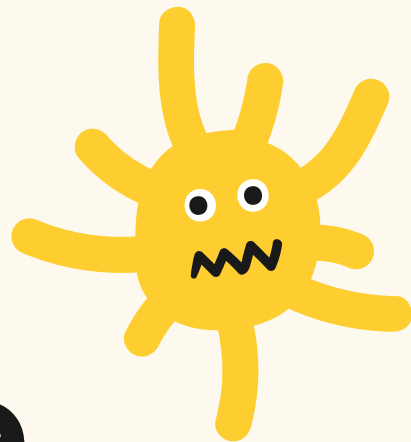
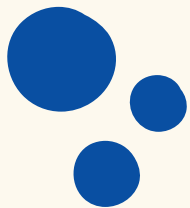
- The **shape of the original population distribution does not matter**. The population could be skewed right, skewed left, or bimodal. As long as the sample size is large enough (a common rule of thumb is $n \geq 30$), the distribution of *sample means* will still be bell-shaped and Normal.
- This theorem is our bridge: it allows us to use the properties of the Normal distribution to make inferences about a population mean, even when we have no idea what the population itself looks like.



Visualizing The Central Limit Theorem

Let's see the CLT in action. We'll start with a population that is heavily **skewed right** (an exponential distribution) and see what happens to the sampling distribution of the mean as we increase our sample size, n .





04

Inference

Confidence Interval, Hypothesis testing, power, type 1&2 error, and sample size



Confidence Interval

- In the real world, we rarely know the true population mean (μ). We use a sample mean ($\bar{x}$) to estimate it. But because samples vary, our estimate is almost never exactly right. A **confidence interval (CI)** gives us a range of plausible values for the true population parameter.
- **Definition:** A confidence interval is an interval computed from sample data that is used to estimate an unknown population parameter.
- **General Formula:** The basic structure for most confidence intervals you'll encounter is: **Point Estimate $\pm$ Margin of Error**
- **Formula for a Mean (with known σ):**

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$ is our sample mean (the point estimate).
- z^* is the **critical value** from the Normal distribution that corresponds to our desired confidence level.
- $\frac{\sigma}{\sqrt{n}}$ is the **standard error** of the mean.



Confidence Interval



- **Common Critical Values (z^*):**
 - For a **90%** confidence interval, $z^*=1.645$.
 - For a **95%** confidence interval, $z^*=1.96$.
 - For a **99%** confidence interval, $z^*=2.576$.



Confidence Interval Interpretation



- The **confidence level** (e.g., 95%) refers to the **long-run success rate of the method**, not to a single interval.
- **Interpretation:** “If we were to take 100 different random samples from this population and construct a 95% CI for each, we would expect about 95 of those intervals to contain the true population mean, μ .”
- It is **incorrect** to say: “There is a 95% probability that the true mean μ is in our calculated interval.” **Why is it wrong?** The true population mean μ is a fixed, unknown number. It doesn’t move. It’s either in our specific interval or it’s not. The randomness comes from the sampling process—it’s the **interval** that changes with every sample.



Confidence Interval Visualization

- This simulation demonstrates the correct interpretation. We'll take 100 samples from a population where we know the true mean is 50. We'll calculate a 95% CI for each sample and see how many of them actually capture the true mean.

```
# Load libraries
library(ggplot2)
library(dplyr)

# Set parameters
set.seed(123)
true_mean <- 50
pop_sd <- 10
sample_size <- 30
n_samples <- 100

# Simulate the 100 samples and their CIs
ci_sim_data <- tibble(
  sample_id = 1:n_samples,
  sample_mean = replicate(n_samples, mean(rnorm(n = sample_size, mean = true_mean, sd = pop_sd))),
  se = pop_sd / sqrt(sample_size),
  lower_bound = sample_mean - 1.96 * se,
  upper_bound = sample_mean + 1.96 * se,
  captures_mean = (lower_bound < true_mean) & (upper_bound > true_mean)
)

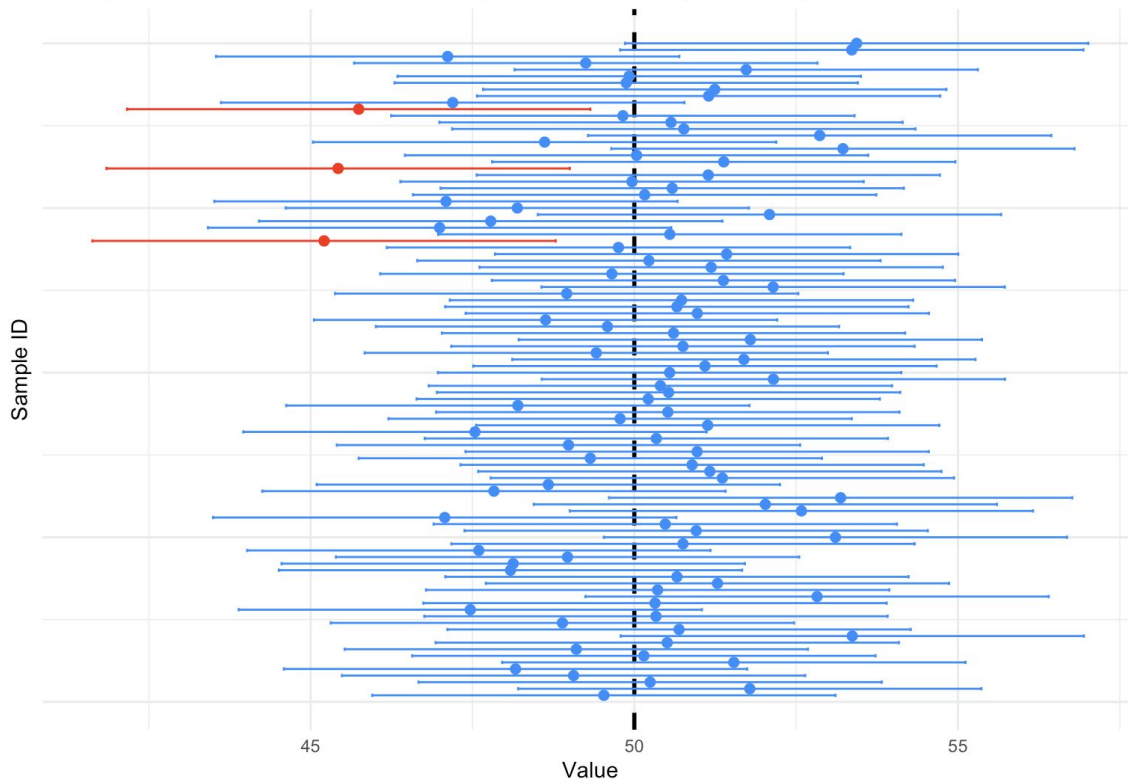
# Plot the CIs
ggplot(ci_sim_data, aes(x = sample_id, y = sample_mean, color = captures_mean)) +
  geom_hline(yintercept = true_mean, color = "black", linetype = "dashed", linewidth = 1) +
  geom_errorbar(aes(ymin = lower_bound, ymax = upper_bound), width = 0.5) +
  geom_point(size = 2) +
  scale_color_manual(values = c("TRUE" = "dodgerblue", "FALSE" = "red")) +
  coord_flip() +
  labs(
    title = "Simulating 100 Confidence Intervals (95% Confidence)",
    subtitle = "We expect about 95 of the 100 intervals to capture the true mean (dashed line).",
    x = "Sample ID",
    y = "Value"
  ) +
  theme_minimal() +
  theme(legend.position = "none", axis.text.y = element_blank(), axis.ticks.y = element_blank())
```



Confidence Interval Visualization

Simulating 100 Confidence Intervals (95% Confidence)

We expect about 95 of the 100 intervals to capture the true mean (dashed line).



Interpretation of a CI



- The **confidence level** (e.g., 95%) refers to the **long-run success rate of the method**, not to a single interval.
- **Correct way to say it:** “We are 95% confident that the **method** used to generate this interval will produce an interval that captures the true population mean.”
- **Another way:** “If we were to take 100 different random samples from this population and construct a 95% CI for each, we would expect about 95 of those intervals to contain the true population mean, μ .”
- It is **incorrect** to say: “There is a 95% probability that the true mean μ is in our calculated interval.” **Why is it wrong?** The true population mean μ is a fixed, unknown number. It doesn't move. It's either in our specific interval or it's not. The randomness comes from the sampling process—it's the **interval** that changes with every sample.





The Logic of Hypothesis Testing

- What is it?
 - Hypothesis testing is a formal procedure for using sample data to decide between two competing claims about a population. It's the statistical version of "innocent until proven guilty."
- The Two Hypotheses
 - Every test starts with two hypotheses:
 1. **The Null Hypothesis (H_0):** This is the "default" assumption, a statement of no effect or no difference. It always includes an equality sign ($=$, $\leq$, or $\geq$). We assume the null is true to begin the test.
 2. **The Alternative Hypothesis (H_a):** This is the research claim we want to find evidence *for*. It's the "interesting" result and never contains an equality sign ($\neq$, $<$, or $>$).



The P-Value and Significant Level



- These two values are at the heart of making a decision in a hypothesis test.
- **P-Value**
 - **Definition:** The p-value is the probability of observing a test statistic as extreme as, or more extreme than, the one you actually got, **assuming the null hypothesis is true**.
 - **Interpretation:**
 - A **small p-value** suggests that your observed data is surprising or rare if the null hypothesis were true, providing evidence *against* the null.
 - A **large p-value** suggests your data is not surprising and is consistent with the null hypothesis.
- **Significance Level (α)**
 - **Definition:** The significance level, α , is a **pre-determined cutoff** for how small a p-value needs to be for us to decide it's "surprising enough" to reject the null hypothesis. The most common value is $\alpha=0.05$.
- **The Decision Rule:**
 - If **p-value $\leq \alpha$** : We reject the null hypothesis (H_0).
 - If **p-value $> \alpha$** : We fail to reject the null hypothesis (H_0).
- **Important:** We never "accept" the null hypothesis. We only ever find that there isn't enough evidence to reject it.





Type I and Type II Errors

- **Type I Error (False Positive):**
 - **What it is:** You **reject H_0** when, in reality, H_0 is true.
 - **Analogy:** A medical test says a healthy person is sick.
 - **Probability:** The probability of a Type I error is equal to the significance level, α .
- **Type II Error (False Negative):**
 - **What it is:** You **fail to reject H_0** when, in reality, H_0 is false .
 - **Analogy:** A medical test says a sick person is healthy. You missed the effect!
 - **Probability:** The probability of a Type II error is denoted by β .

	Reality: H_0 is True	Reality: H_a is True	
Decision: Fail to Reject H_0	Correct Decision	Type II Error (β)	
Decision: Reject H_0	Type I Error (α)	Correct Decision (Power)	





Statistical Power & Sample Size

- **Statistical Power**
 - **Definition:** Power is the probability of correctly rejecting a false null hypothesis. It's the probability that your test will detect a real effect if one truly exists.
 - **Formula:** Power is the complement of a Type II error: **Power = $1 - \beta$** .
 - **Goal:** We want studies to have high power, typically 80% or more.
- **Factors that Increase Power**
 1. **Increase the Sample Size (n):** This is the most common way. More data reduces error and makes the signal clearer .
 2. **Increase the Effect Size:** A larger true difference is easier to detect .
 3. **Increase the Significance Level (α):** This makes it easier to reject H_0 , but it also increases the risk of a Type I error.



Statistical Power & Sample Size



- Sample Size Calculation

Before a study, we calculate the required sample size to ensure it has enough **power** to detect a meaningful effect. To do this, we must specify our desired α , power, the estimated population variability (σ), and the smallest **effect size** that we care about .



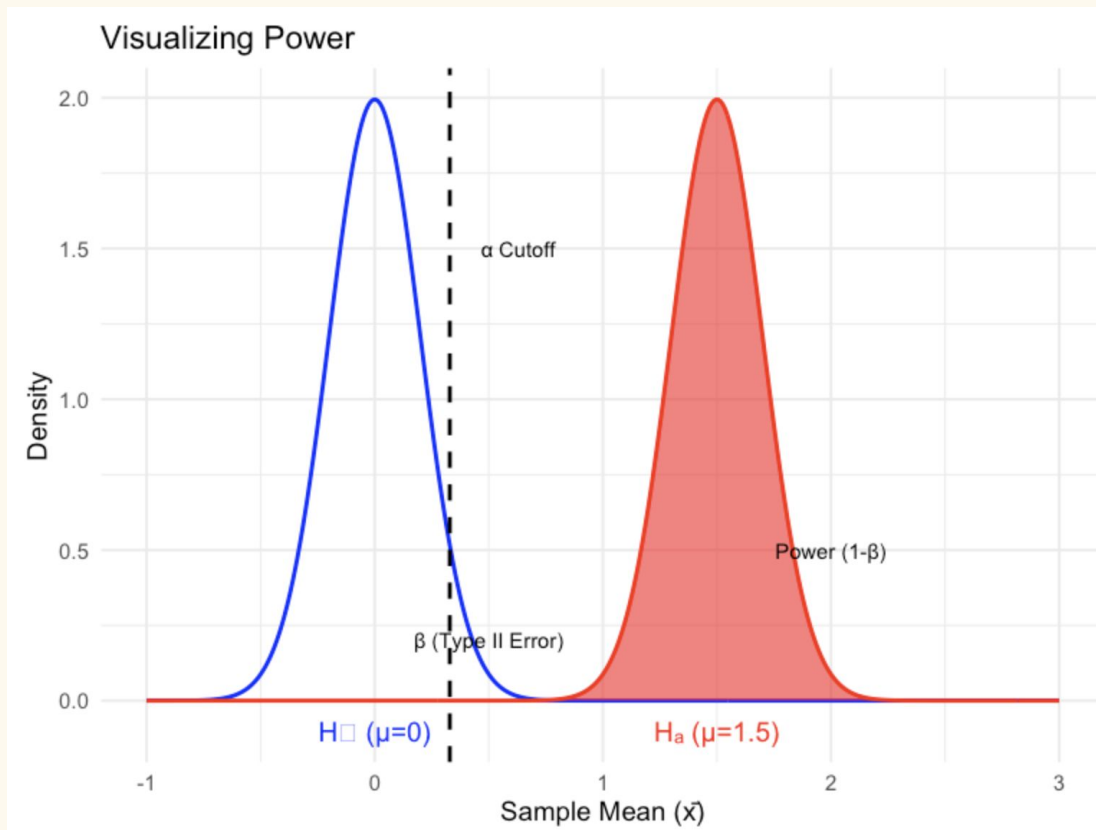


Visualizing Power

- The relationship between H_0 , H_a , β , and Power can be visualized with two overlapping distributions.
- The **blue curve** represents the sampling distribution if the **null hypothesis is true** ($\mu = \mu_0$).
- The **red curve** represents the sampling distribution if the **alternative hypothesis is true** ($\mu = \mu_a$).
- The **black line** is our decision cutoff, determined by α .
- **β** is the area under the red (H_a) curve that falls in the "fail to reject" region. It's the chance we'll mistake a sample from the red world as coming from the blue world.
- **Power** is the area under the red (H_a) curve that falls in the "reject" region. It's the chance we'll correctly identify that our sample came from the red world.



Visualizing Power



Sample Size Calculation



- **Why Calculate Sample Size?**
 - Before conducting an expensive study, researchers calculate the required sample size to ensure their study has **enough power** to detect an effect of a certain size, if it truly exists. It's a crucial step in study design to avoid wasting resources or conducting a study that is doomed to fail (i.e., has low power).
- **What You Need to Know**
 - To calculate the sample size for a simple Z-test for a mean, you must specify four quantities in advance:
 1. **Significance Level (α):** The risk of a Type I error you're willing to accept (e.g., 0.05).
 2. **Desired Power ($1-\beta$):** The chance of detecting a true effect you want to have (e.g., 80% or 0.80).
 3. **Standard Deviation (σ):** An estimate of the population standard deviation, often from previous research.
 4. **Effect Size:** The smallest difference between the null mean (μ_0) and the alternative mean (μ_a) that you consider scientifically or practically meaningful.



Sample Size Calculation



Formula for a One-Sided Test for a Mean

The formula combines these components to solve for n:

$$n = \left(\frac{(z_{\alpha} + z_{\beta})\sigma}{(\mu_a - \mu_0)} \right)^2$$

- z_{α} is the z-score for your significance level (e.g., for $\alpha=0.05$ one-sided, $z_{\alpha}=1.645$).
- z_{β} is the z-score for your desired power (e.g., for 80% power, $\beta=0.20$, and $z_{\beta}=0.84$).



Sample Size Calculation

Example Question & Calculation

You want to test if a new diet reduces cholesterol from a known mean of 211 mg/100ml. You estimate $\sigma=46$. How many people do you need to have 95% power to detect a true mean of 180 mg/100ml at an $\alpha=0.01$ significance level?

```
```{r}
--- Step 1: Define all the parameters from the problem ---
alpha <- 0.01
power <- 0.95
beta <- 1 - power # Beta is the complement of power

sigma <- 46
mu_null <- 211
mu_alt <- 180
effect_size <- abs(mu_alt - mu_null)

--- Step 2: Find the corresponding Z-scores ---
For a one-sided test (since we are testing for a *reduction*)
z_alpha <- qnorm(1 - alpha)
z_beta <- qnorm(power) # or qnorm(1-beta)

--- Step 3: Apply the formula ---
n_calculated <- ((z_alpha + z_beta) * sigma / effect_size)^2

--- Step 4: Round up to the nearest whole number ---
n_required <- ceiling(n_calculated)

--- Print the results in a clear format ---
cat("--- Sample Size Calculation ---\n")
cat("Significance Level (alpha):", alpha, "\n")
cat("Desired Power (1-beta):", power, "\n")
cat("Effect Size ($\mu_a - \mu_0$):", effect_size, "\n")
cat("Estimated Sigma (σ):", sigma, "\n\n")
cat("Z-score for alpha (z_alpha):", round(z_alpha, 3), "\n")
cat("Z-score for beta (z_beta):", round(z_beta, 3), "\n\n")
cat("Calculated sample size (n):", n_calculated, "\n")
cat("Required sample size (rounded up):", n_required, "\n")
```
```

--- Sample Size Calculation ---
Significance Level (alpha): 0.01
Desired Power (1-beta): 0.95
Effect Size ($\mu_a - \mu_0$): 31
Estimated Sigma (σ): 46

Z-score for alpha (z_alpha): 2.326
Z-score for beta (z_beta): 1.645

Calculated sample size (n): 34.72451
Required sample size (rounded up): 35

